

Equation Sample 1

Dataset-expansion sample

1 Derivation

The following display is drawn from a source-backed image-to-Latex benchmark and reproduced verbatim.

$$\begin{aligned}
\alpha_3 &\stackrel{(i)}{\leq} \sum_{s \in \mathcal{S}, a \in \mathcal{A}} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \sqrt{\frac{C_{\text{clipped}}^* \log \frac{N}{\delta}}{N \min \left\{ d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho), \frac{1}{S(A+B)} \right\}}} \text{Var}_{P_{s, a, \nu_0(s)}}(V_{\text{pe}}^-) \\
&\leq \sqrt{\frac{C_{\text{clipped}}^* \log \frac{N}{\delta}}{N}} \sum_{s \in \mathcal{S}, a \in \mathcal{A}} \sqrt{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s, a, \nu_0(s)}}(V_{\text{pe}}^-)} \\
&\quad + \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \sum_{s \in \mathcal{S}, a \in \mathcal{A}} \sqrt{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho)} \cdot \sqrt{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s, a, \nu_0(s)}}(V_{\text{pe}}^-)} \\
&\stackrel{(ii)}{\leq} \sqrt{\frac{C_{\text{clipped}}^*}{N} \log \frac{N}{\delta}} \cdot \sqrt{SA} \cdot \sqrt{\sum_{s \in \mathcal{S}, a \in \mathcal{A}} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s, a, \nu_0(s)}}(V_{\text{pe}}^-)} \\
&\quad + \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \left[\sum_{s \in \mathcal{S}, a \in \mathcal{A}} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \right] \sqrt{\sum_{s \in \mathcal{S}, a \in \mathcal{A}} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s, a, \nu_0(s)}}(V_{\text{pe}}^-)} \\
&\leq 2 \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \sqrt{\sum_{s \in \mathcal{S}, a \in \mathcal{A}} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s, a, \nu_0(s)}}(V_{\text{pe}}^-)} \\
&\stackrel{(iii)}{=} 2 \sqrt{\frac{C_{\text{clipped}}^* S(A+B)}{N} \log \frac{N}{\delta}} \sqrt{\sum_{s \in \mathcal{S}} d^{\mu^*, \nu_0}(s; \rho) \mathbb{E}_{a \sim \mu^*(s), b \sim \nu_0(s)} [\text{Var}_{P_{s, a, b}}(V_{\text{pe}}^-)]} \\
&\stackrel{(iv)}{\leq} 2 \sqrt{\frac{C_{\text{clipped}}^* S(A+B)}{N} \log \frac{N}{\delta}} \sqrt{\sum_{s \in \mathcal{S}} d^{\mu^*, \nu_0}(s; \rho) \text{Var}_{P_s^{\mu^*, \nu_0}}(V_{\text{pe}}^-)}.
\end{aligned}$$